

Operations Management

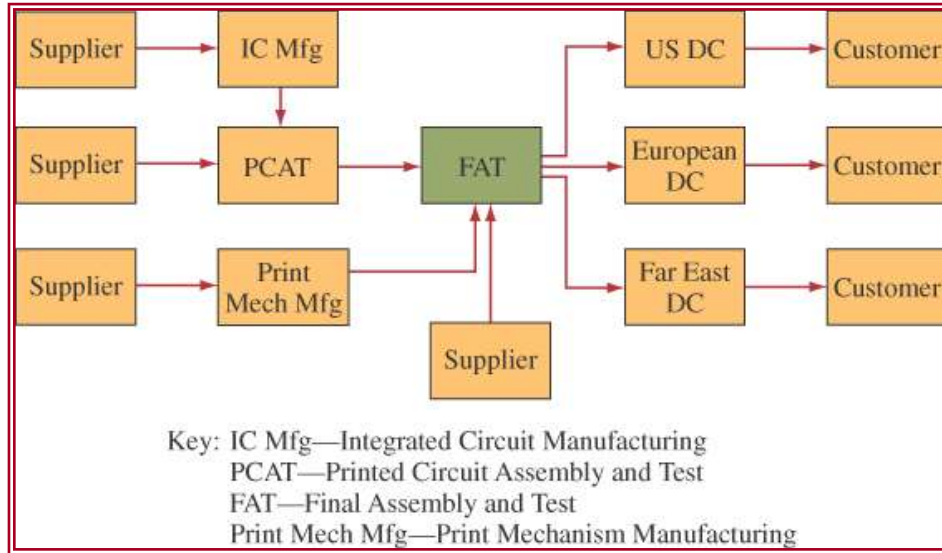
Session 8: Quick response, Distributions & Logistics Class Wrap-Up

Outline:

- Production Postponement
- Quick Response Initiatives
- Zara
- Supply Chain Logistics and Distribution Management
 - Warehousing and Inventory Management
 - Square root law of inventory pooling
 - Last mile logistics
 - Traveling Salesman Problem (TSP)
 - Square root law of routing
- Course Wrap-Up

Postponement

HP DeskJet Supply Chain



Problem:

- Inventory Crisis in Europe
- Some countries with stockouts
- Some countries with excess inventory
- Changing market conditions
- Poor Service Level

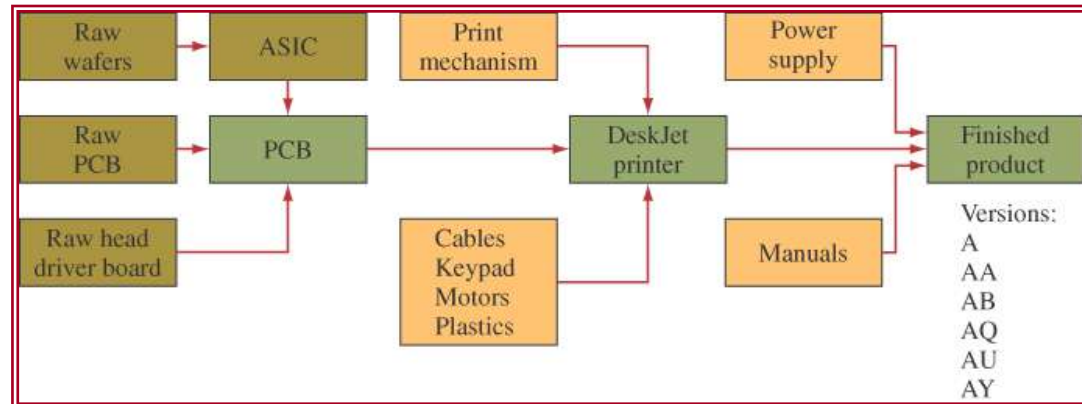
Facts:

- Late 80's one of HP most popular product
- Centralized production in Vancouver
- More than 6 weeks leadtime to Europe

More Facts:

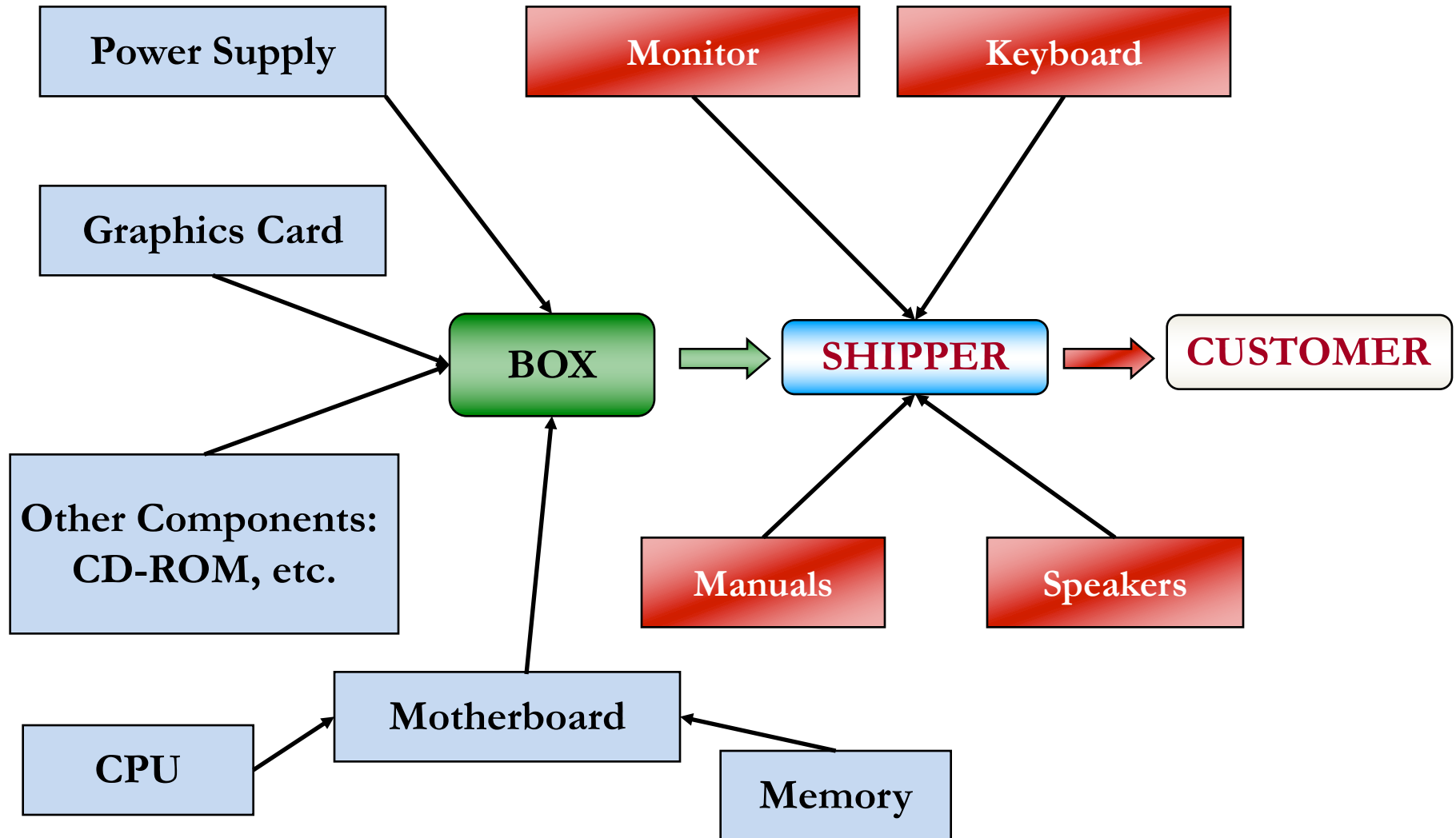
- Six different versions for Europe
- Difference in Language and Power Supply
- Product are “**localized**” in Vancouver

HP DeskJet Bill of Materials



Other examples of postponement

Other examples of postponement



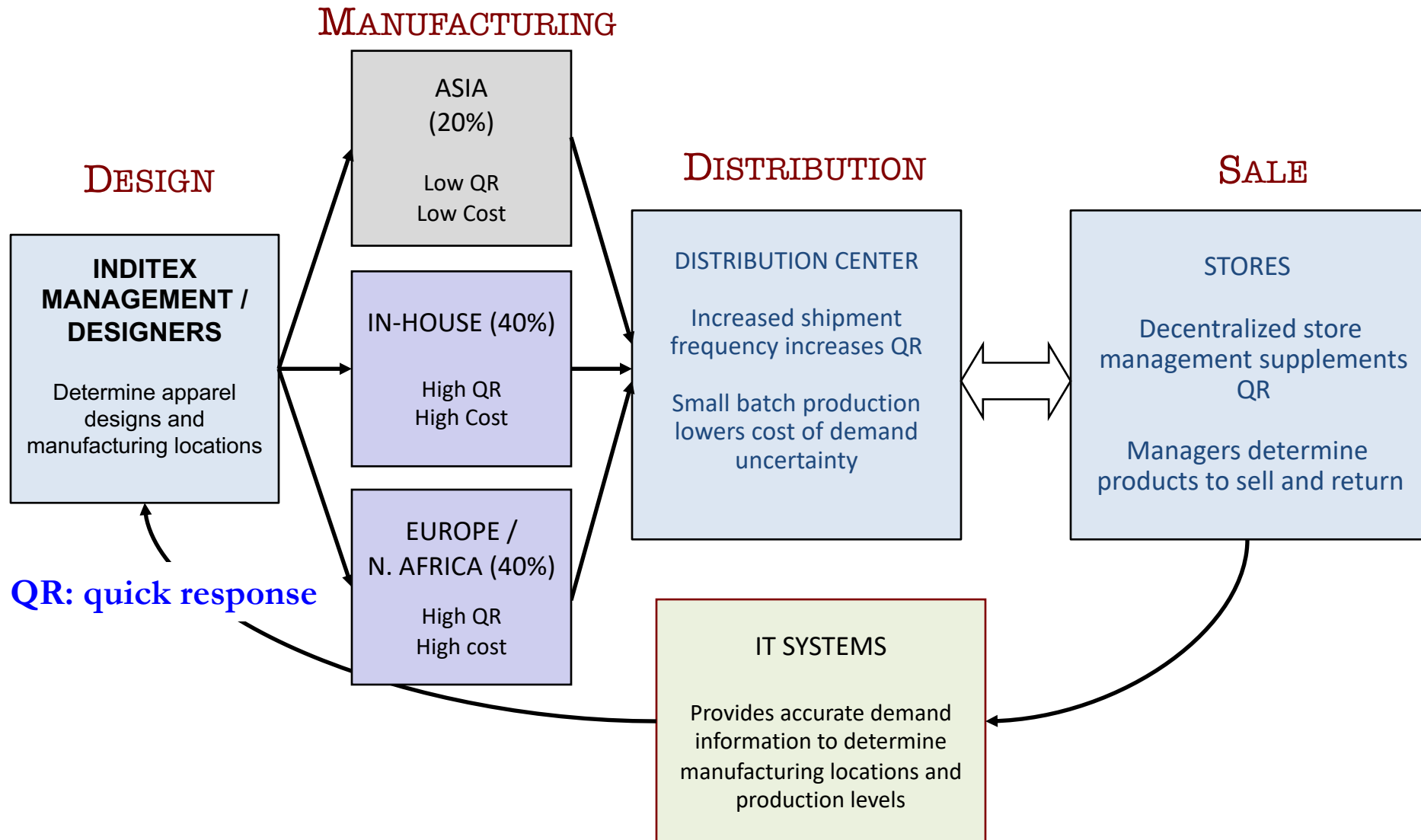
Quick Response ZARA



- ✓ Sales growth at 20%/yr since '90s.
- ✓ Net Profit margin at 10%
- ✓ NPI once a week,
- ✓ 12K new product/year
- ✓ 23-30 days of final design to sales cycle.
- ✓ A typical customer visits a Zara store more than 15 times a year.

Zara Value Proposition

Zara Supply Chain: Fast Fashion

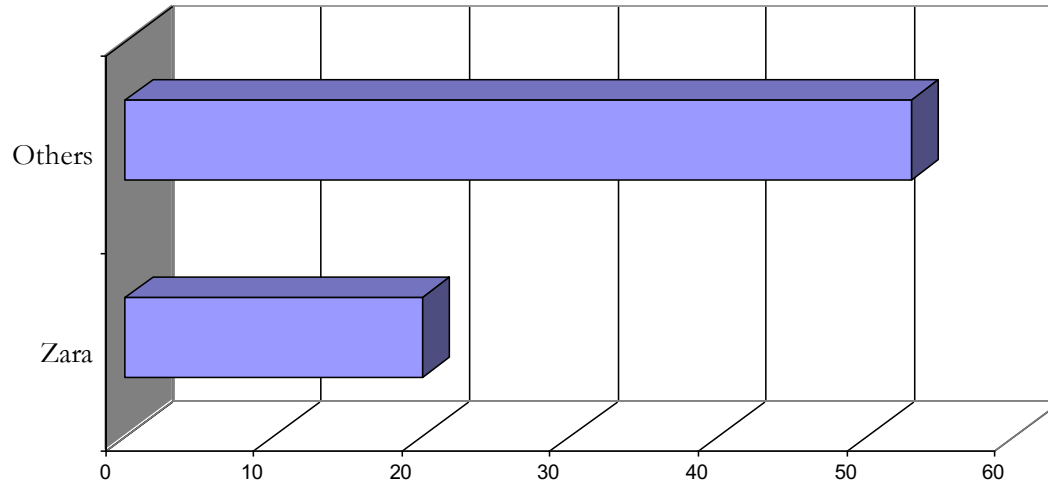


ZARA's Value Chain

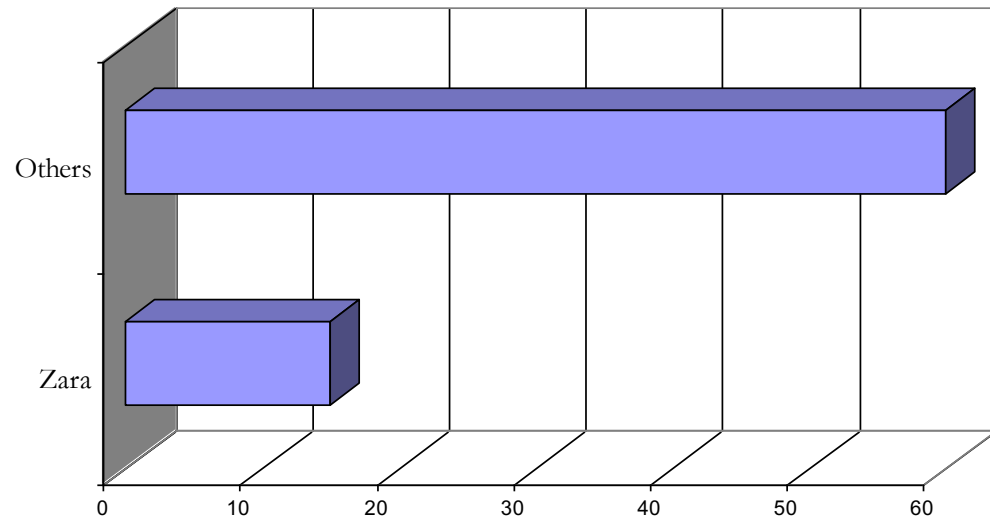
Process Step	Design	Rationales
Material Purchasing	In-House	Scale econ., Design Control
Cutting	In-House	Automation, Scale
Sewing	Outsource	Labor-Intensive, Flexibility
Finishing, Packing	In-House	Automation, Flexibility, Control
Distribution	In-House	Automation, Speed, Control
Retail Stores	Owned	Control, Company Image

ZARA: JIT production

% of material commitment 6 months prior to season

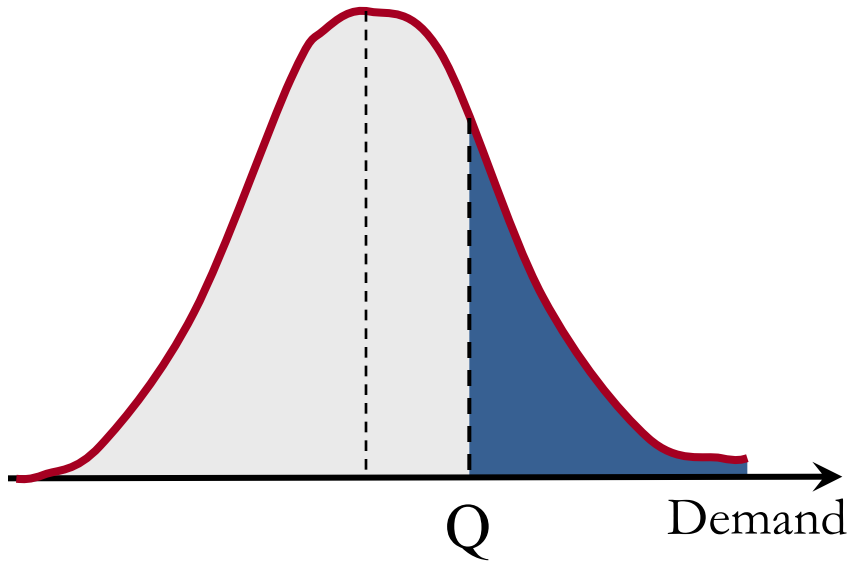


% of production at start of season

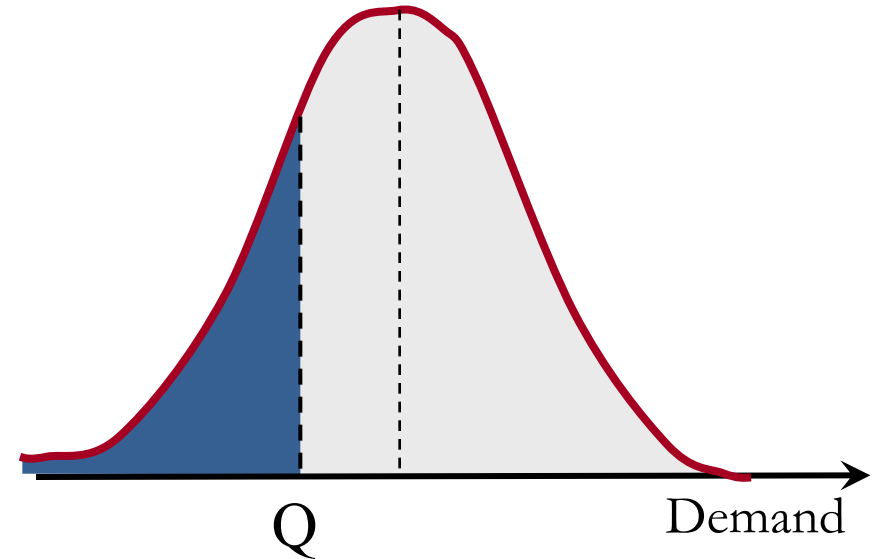


Retailer	Inventory Turnover Rate (per year)
Zara	12.0
H&M	3.5
Uniqlo	4.5
Gap Inc.	4.0

The Fashion Gamble



- Lower cost of excess through use of markdowns & sales
- High cost of shortage with lost sales



- High cost of excess with no markdowns & sales
- Low cost of shortage with rolling mix NPI

How is Zara Changing the odds (reduce uncertainty)?

Zara: QR Cost/Benefit Analysis

Learning Customers' Preferences

Two Contrasting Examples of Fast-Fashion

Metric	Zara (Spain)	H&M (Sweden)	The Gap (U.S.A.)
Vertical integration	Fully integrated; subcontracts cutting, sewing, and shipping	Controls every link in the chain but does not own factories	From design to store but outsources production
No. of stores worldwide	1,759	3,777	3,592
No. of countries	68	28	21
Distribution of stores – main locations	13% North Europe 60% South Europe 7% East Europe 8% Latin America 5% Asia Pacific	63.5% North Europe 19% South Europe 5% East Europe 12% North America	9% United Kingdom 79% North America 7% Japan
Sourcing – main suppliers	34% Asia 50% Spain & prox. 14% Rest of Europe	> 60% Asia < 40% Europe	97% outside U.S.A.
Assortment composition	40% Basic 60% Fashion	> 70% Basic < 30% Fashion	99% Basic
Lead times – dual supply chain (SC)	Efficient SC: 6 months Responsive SC: 2–5 weeks	Efficient SC: 6 months Responsive SC: 3–6 weeks	Efficient SC: 9 months
Refresh fashion items	Twice a week	Daily	Occasionally
Pricing	Overall higher than H&M (especially outside Spain)	Lowest among fast fashion	Comparable to Zara, if not higher
Marketing expenditure	0.3% of revenues	3–4% of revenues	Comparable to H&M

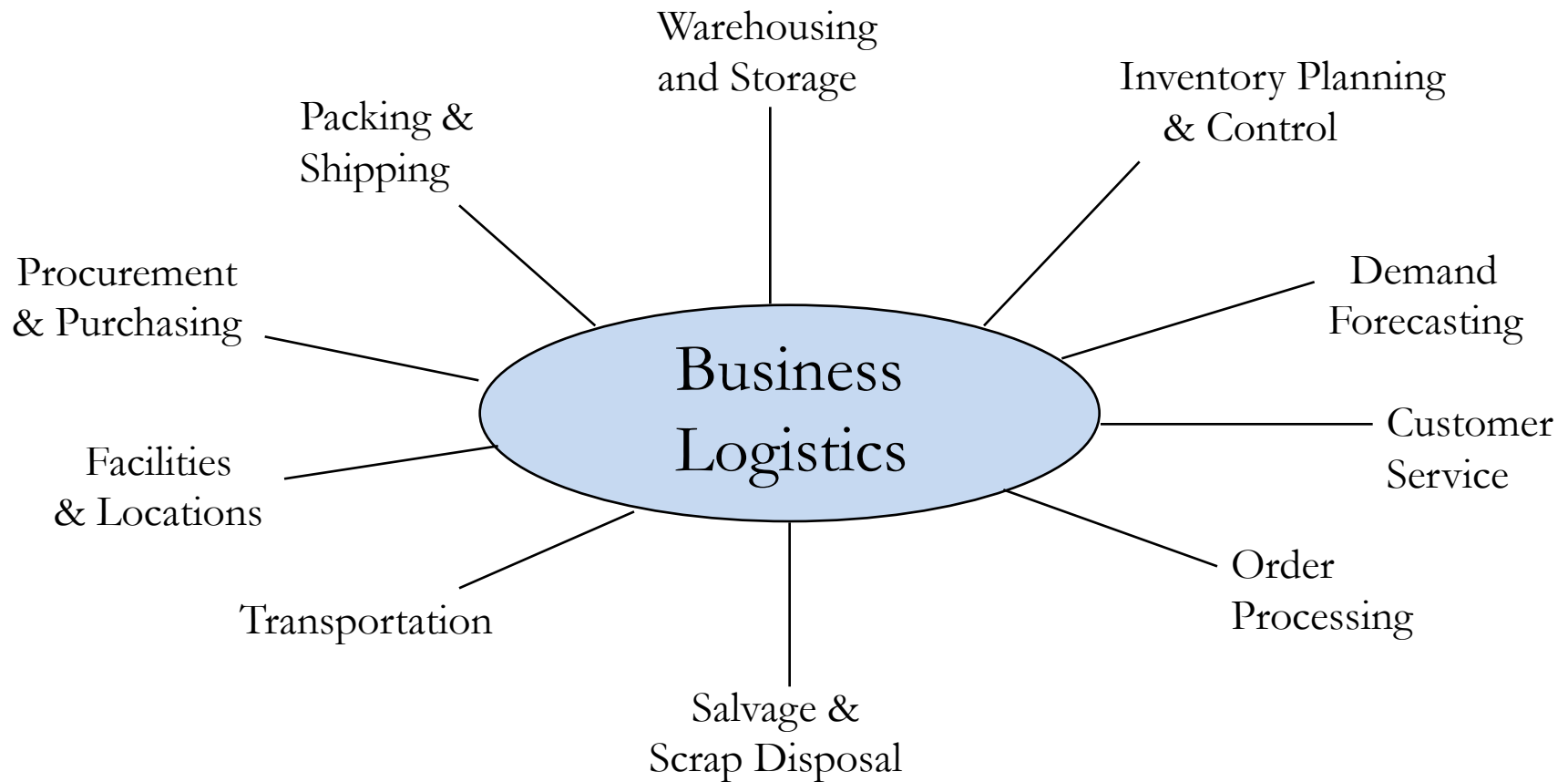
AI-Support Fashion Design

AI Tailoring: A Framework for Predicting Fashion Popularity

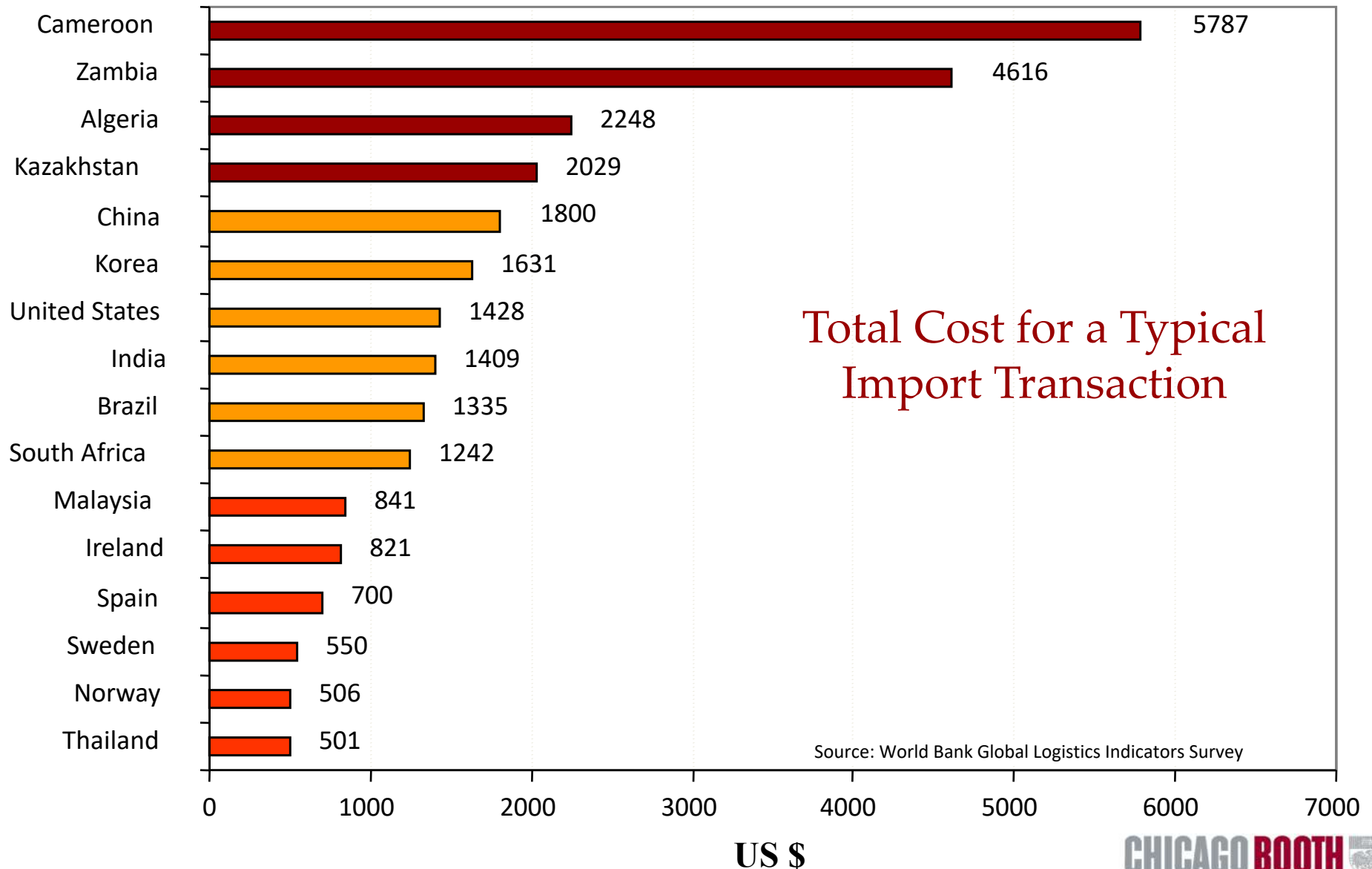


Source: AI Tailoring: Evaluating Influence of Image Features on Fashion Product Popularity [\[Link\]](#)

Logistics

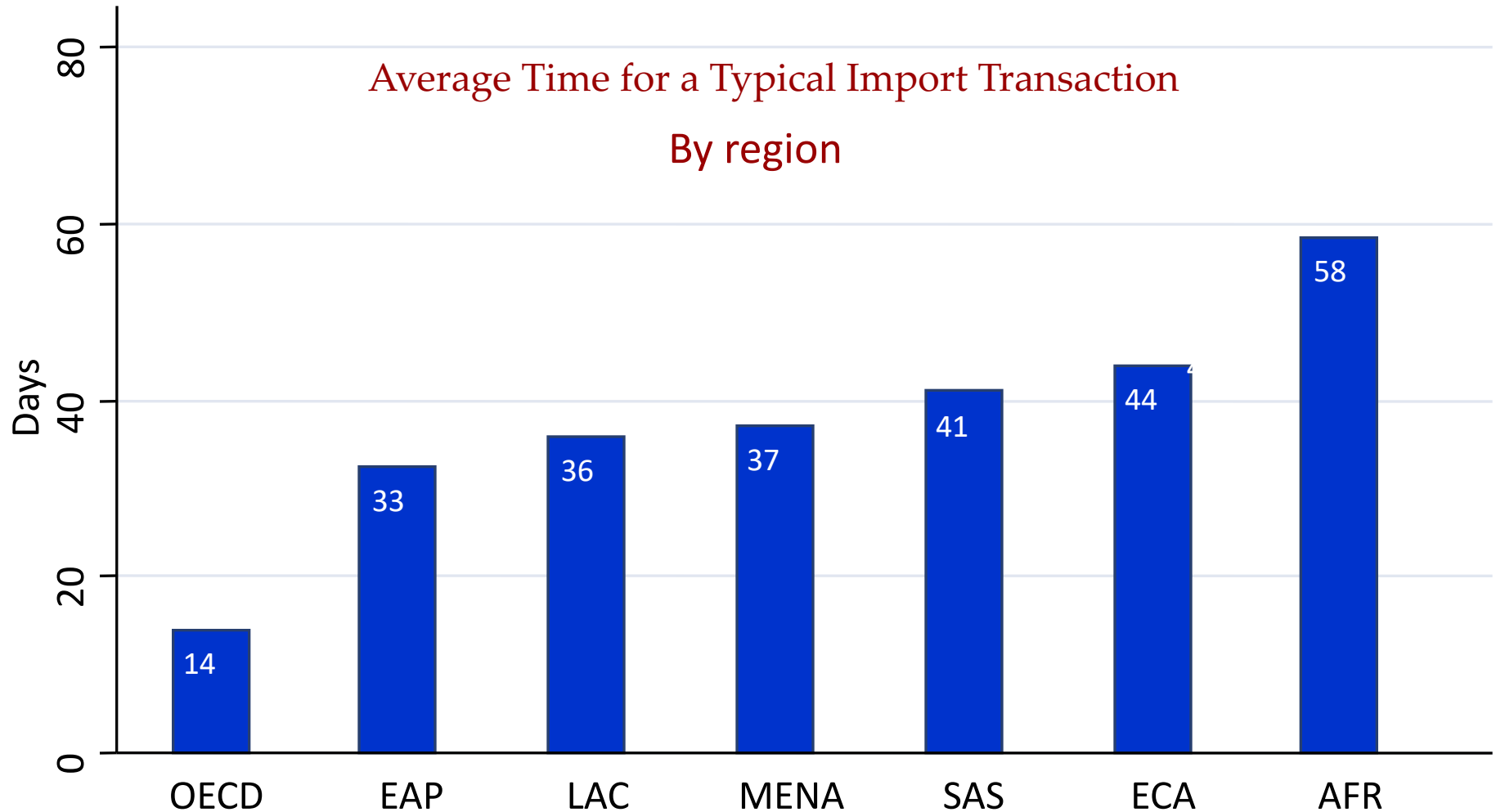


Global Logistics



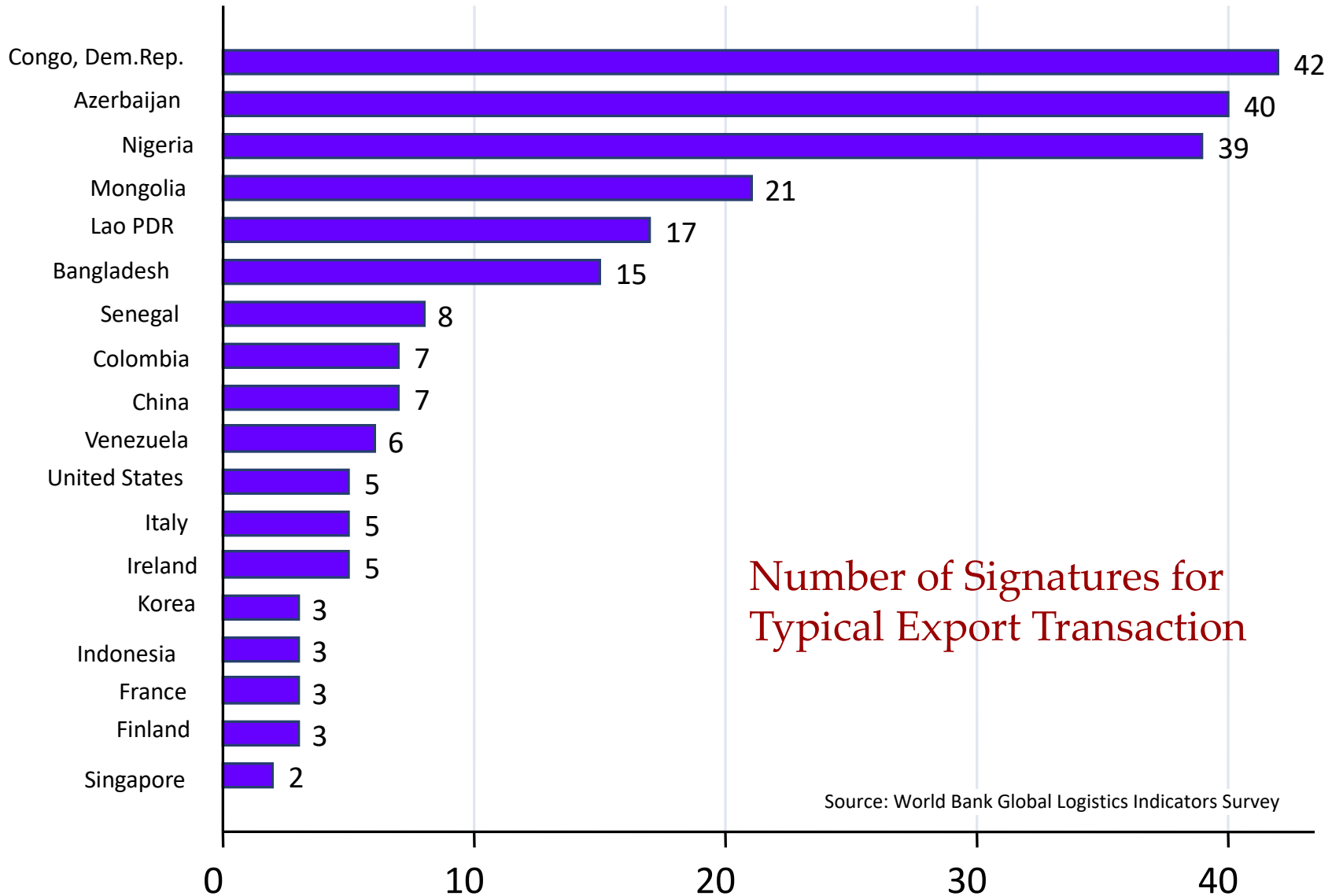
Global Logistics

Average Time for a Typical Import Transaction
By region



Source: World Bank Global Logistics Indicators Survey

Global Logistics

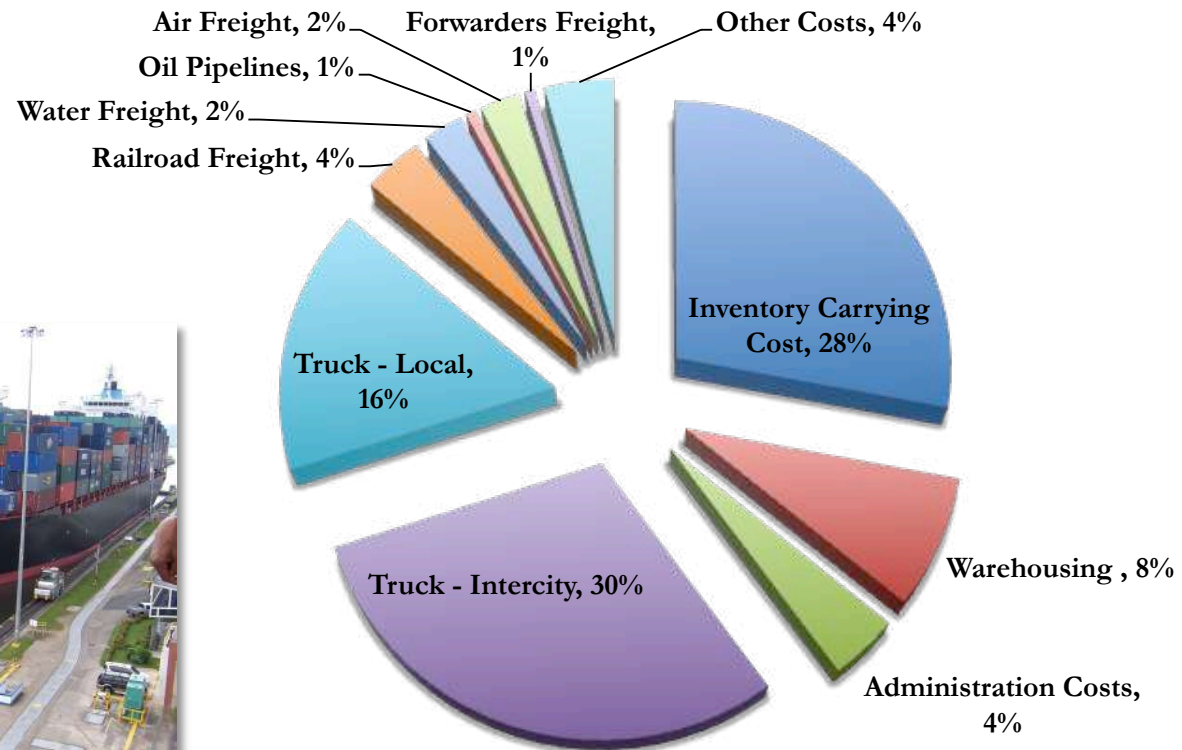


Number of Signatures for
Typical Export Transaction

Source: World Bank Global Logistics Indicators Survey

Component of Logistics Costs

Total US Logistics Costs 2023 (US\$ 2.4 Trillion, 9% GDP)

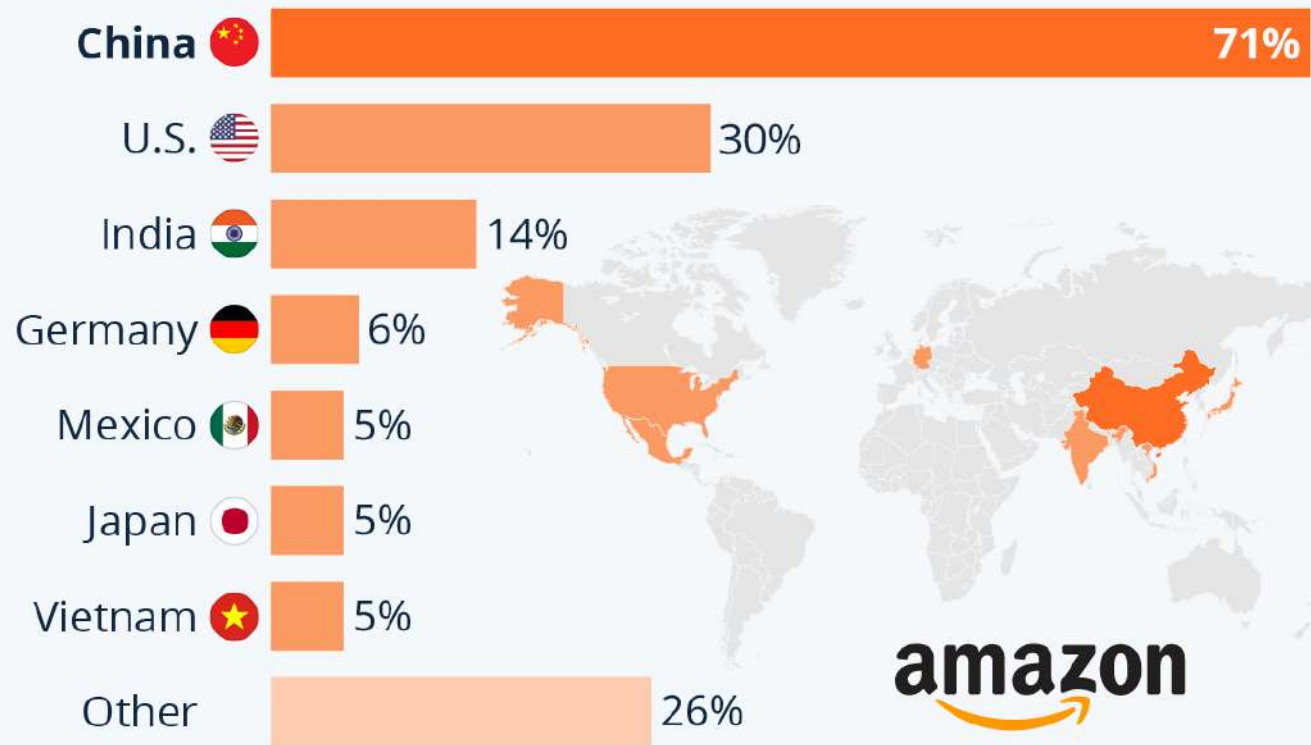


Transportation Cost + Inventory Costs = 84%

Amazon Inbound Logistics

Made in China, Sold on Amazon

Share of items sold on Amazon, by country of origin*



* The retailers surveyed listed several countries as source locations, which explains why the total share is above 100%.

Amazon Outbound Logistics

Amazon's Escalating Logistics Costs

Amazon's fulfillment and shipping costs in total and as a percentage of net sales



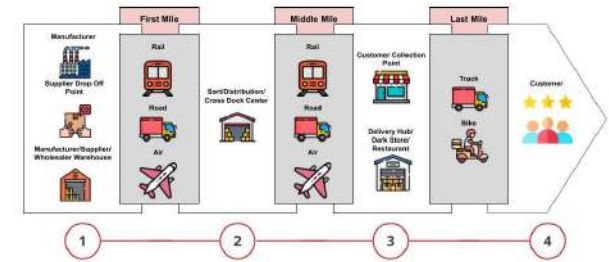
* costs incurred in operating and staffing fulfillment centers, customer service centers and physical stores as well as payment processing costs

Source: Amazon



statista

Logistics Management



Inventory strategy

- ✧ Warehousing
- ✧ Pooling strategies
 - ✓ JungleBooks.com
- ✧ Push-pull strategies
- ✧ Postponement and Mass Customization

Distribution strategy

- ✧ Fundamentals of a delivery business
 - ✓ Tesco/Webvan
- ✧ Last-mile e-commerce
 - order picking
 - delivery routing
 - Traveling Salesman Problem (TSP)
 - density

Types of Distribution

DIRECT SHIPPING: Goods delivered directly from “factory”

- **Pros:** Reduce pipeline inventory and holding costs
- **Cons:** Long leadtimes and high delivery costs

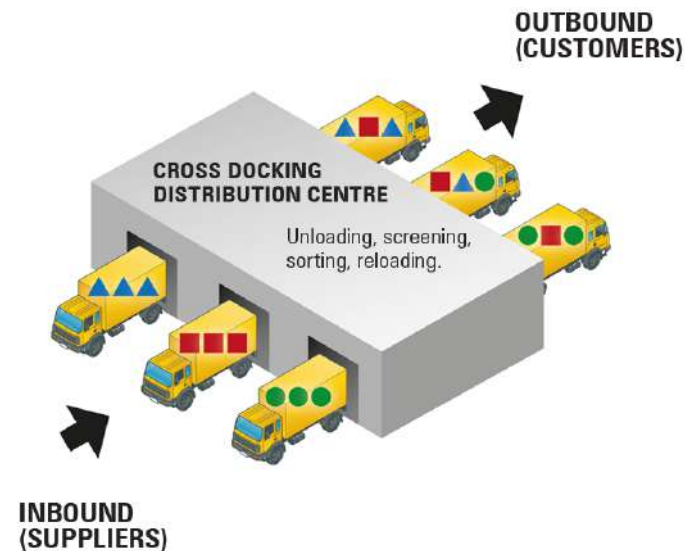
WAREHOUSING: Goods stored at an intermediate location

- **Pros:** Risk Pooling, Transportation (large inbound vs small outbound truck), Mix and Assortment (load consolidation), Break Bulking.
- **Cons:** Inventory can build up in the system, longer cycle times

CROSS DOCKING: Goods transferred at an intermediate location

- **Pros:** Combine the benefits of direct shipping and warehousing
- **Cons:** Complex, difficult to implement

warehousing without storage



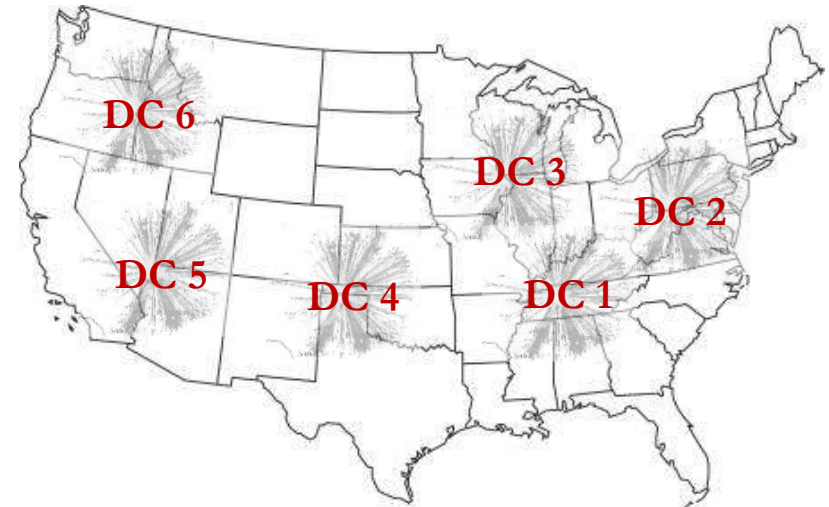
Centralize or Localize? The DC Dilemma

Current DC Operations



PROS of Proposed Ops

Proposed DC Operations



CONS of Proposed Ops

Amazon

Snapshot Data Amazon.com

	Off-Peak Season				Peak Season
	Day 1	Day 2	Day 3	Day 4	Day 5
Total Orders	869K	925K	918K	956K	1.55M
Total SKUs	411K	385K	388K	406K	526K
Single Order	64%	65%	66%	65%	56%
Split Orders	3.9%	3.9%	3.7%	3.6%	6.4%

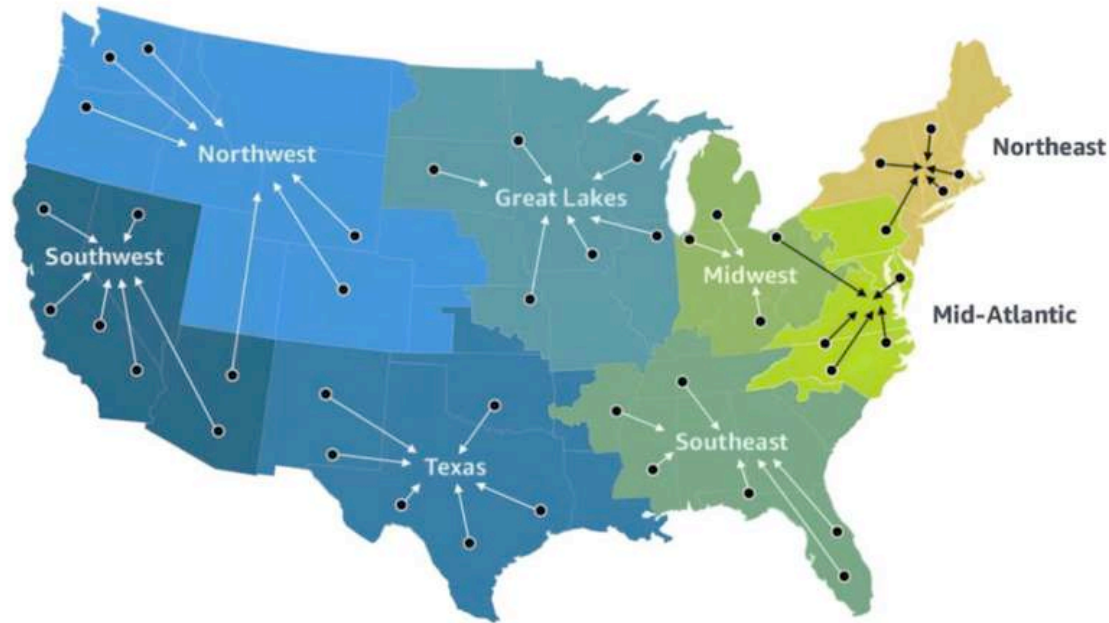
Split orders cost Amazon more than \$200,000 a day.

Another E-commerce Problem: Returns

For online purchases, the average rate of return in 2023 was at nearly 25%, up from 18% in 2020.
The cost of a return can be 1.5–2 times the original delivery cost

Amazon's Regionalization

Figure 3. Map Showing the Approximate Structure of the Eight-Region Design



Notes. The dots represent FCs assigned to serve each region. Boundaries and FC locations are approximate and intended to be only directionally representative.

Source: *Regionalize and Scale: Amazon's Fulfillment Network Design for Faster and Cheaper Delivery* [\[Link\]](#)

The National-to-Regional Shift

- **15% reduction** in average distance between sites and customers.
- **12% fewer** middle-mile touchpoints (simpler flow, fewer handoffs).
- **In-region fulfillment increased** from 62% to 76%.
- **Cost-to-serve per unit fell** for the first time since 2018, including **>\$0.45 per-unit savings** in the U.S.
- **Faster delivery at scale:** 9B+ items delivered same/next day globally in 2024.

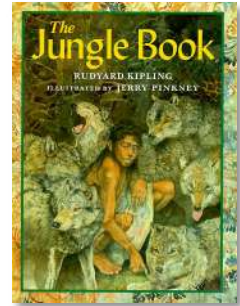
Example: Decentralized Inventory JB.com

	SoftCover I
Number of Titles	50,000
Unit Cost	\$2.00
Lead Time	2
Mean Demand	25
Std Dev	5
Fed Ex Charges	\$6.00

Order cost: $s = \$50$

Standard UPS cost: \$2

Interest rate: $i = 20\%$ annually



replenishment
from suppliers

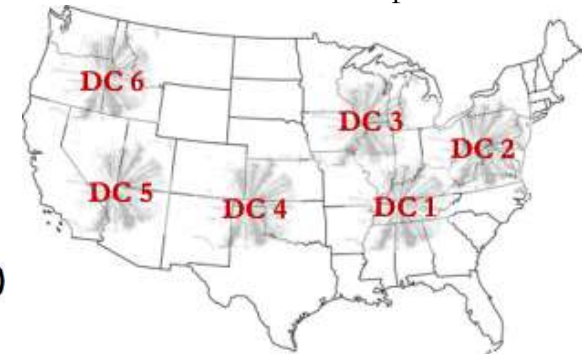
$L = 2$ weeks

Regional
DC

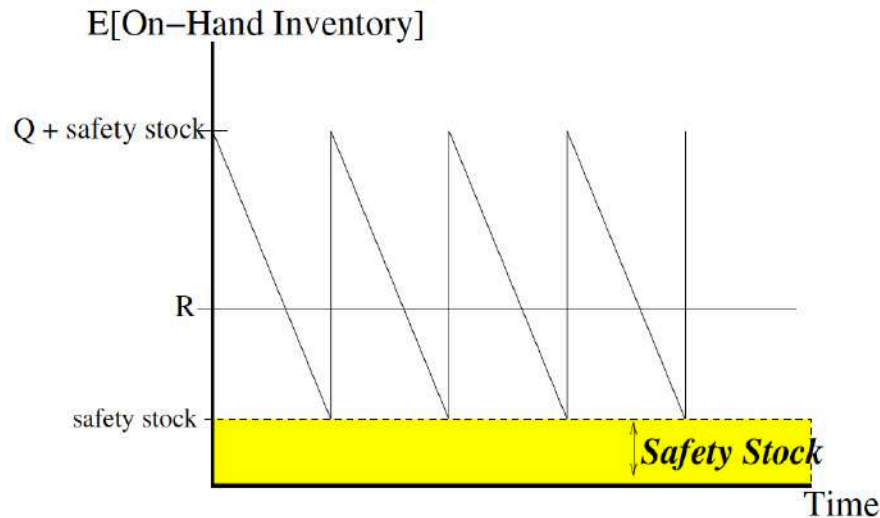
customer
demand

$N(25, 5^2)$

Decentralized DC Operations



Cycle and safety stocks (Q,R)



-) Order Quantity: $Q^* = \sqrt{\frac{2 \lambda s}{h}}$

-) Reorder Point: $R^* = \lambda L + Z_{SL} \sigma \sqrt{L}$

-) Average Inventory: $INV = \frac{1}{2} Q^* + Z_{SL} \sigma \sqrt{L}$

Cycle Stock: Q

$s = \$50$

$\lambda = 25 \times 52 = 1300$ items per year

$h = i \times c = 20\% \times \$2 = \$0.4$ per item per year

$$\left. \begin{array}{l} s = \$50 \\ \lambda = 1300 \\ h = \$0.4 \end{array} \right\} \Rightarrow Q^* = \sqrt{\frac{2 \times 1,300 \times 50}{0.4}} = 570 \text{ units}$$

How to determine SL?

$C_s = \text{cost to FedEx vs. UPS} = \$6 - \$2 = \4

$C_e = h \times Q^* / \lambda = 0.4 \times 570 / 1,300 = \0.175

$$\left. \begin{array}{l} C_s = \$4 \\ C_e = \$0.175 \end{array} \right\} \Rightarrow \begin{array}{l} SL = \$4 / (\$4 + \$0.175) = 0.958 \\ Z_{SL} = 1.73 \end{array}$$

Reorder Point: R

$R^* = 25 \times 2 + 1.73 \times 5 \times \sqrt{2} = 62.37$ items

Avg. INV = $570/2 + 1.73 \times 5 \times \sqrt{2} = 297$ items

Inventory costs, decentralized

Holding Costs: Avg. INV for a SKU = 297 units
Holding cost for one SKU = $h \times \text{INV} = 0.4 \times 297 = \119 per SKU per year

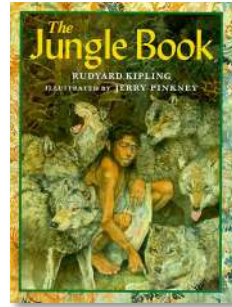
Ordering Costs: Number of orders = $\lambda / Q^* = 1,300/570 = 2.28$ orders per year
Ordering cost for one SKU = $s \times \lambda / Q^* = 50 \times 2.28 = \114 per SKU per year

Total cost per SKU at a single DC = $119 + 114 = \$233$ per year

Total cost per SKU across all six DC = $233 \times 6 = \$1,398$ per year

Total costs for Soft Cover I across all six DC = $1,398 \times 50,000 = \$69,900,000$ per year

Centralized Inventory: Soft Cover I



SoftCover I	
Number of Titles	50,000
Unit Cost	\$2.00
Lead Time	2
Mean Demand	25
Std Dev	5
Fed Ex Charges	\$6.00

Order cost: $s = \$50$

Standard UPS cost: \$2

Interest rate: $i = 20\%$ annually

replenishment
from suppliers

$L = 2$ weeks

Consolidated
DC

customer
demand

$N(6 \cdot 25, 6 \cdot 5^2)$

$\lambda = 150$

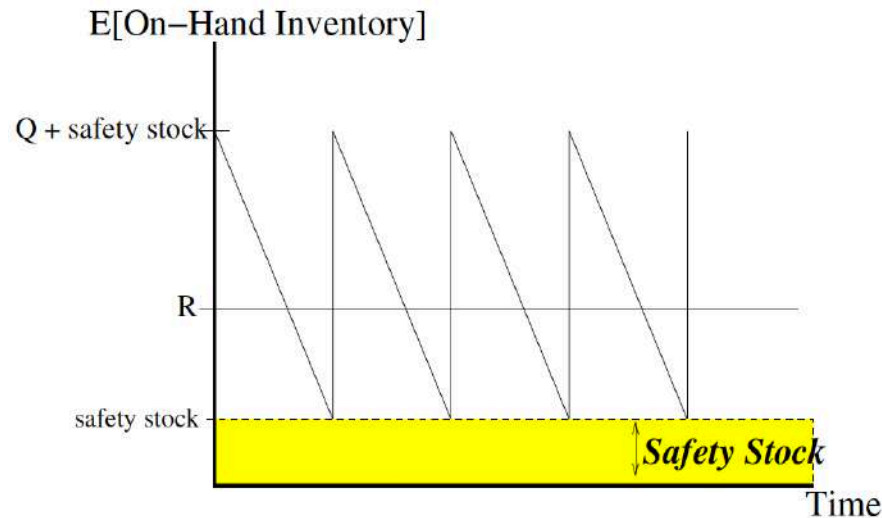
$\sigma = 5\sqrt{6} = 12.25$

Centralized DC Operations



- * assume demand in each location is independent
- * assume same service level

Centralized Inventory: Soft Cover I



-) Order Quantity: $Q^* = \sqrt{\frac{2 \lambda s}{h}}$

-) Reorder Point: $R^* = \lambda L + Z_{SL} \sigma \sqrt{L}$

-) Average Inventory: $INV = \frac{1}{2} Q^* + Z_{SL} \sigma \sqrt{L}$

Cycle Stock:

$$Q^* = \sqrt{\frac{2 \times 7,800 \times 50}{0.4}} \approx 1,397 \text{ units}$$

Reorder Point:

$$R^* = 150 \times 2 + 1.73 \times 12.25 \times \sqrt{2} = 330 \text{ items}$$

$$\text{Avg. INV} = 1,397/2 + 1.73 \times 12.25 \times \sqrt{2} = 728 \text{ items}$$

Holding cost for one SKU = $h \times INV = 0.4 \times 728 = \291 per SKU per year

Ordering cost for one SKU = $s \times \lambda / Q^* = 50 \times 7,800/1,397 = \279 per SKU per year

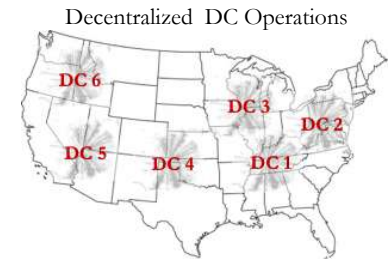
Total cost per SKU = $291 + 279 = \$570$ per SKU per year

Total costs for Soft Cover I = $570 \times 50,000 = \$28,500,000$ per year

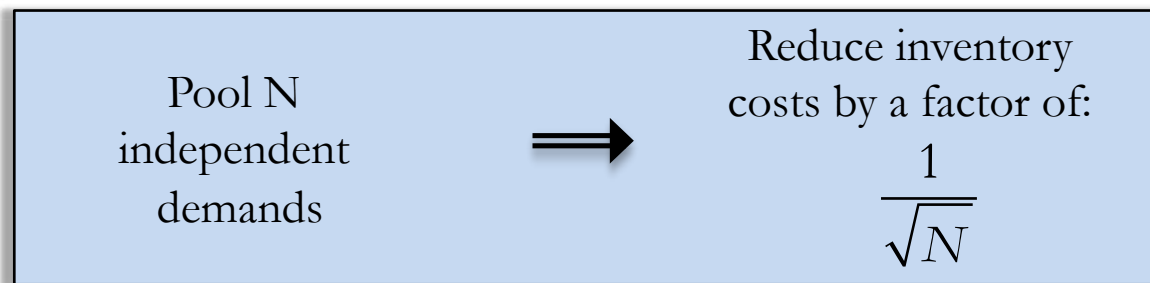
In Summary:



$$\$28,500,000 = \frac{1}{\sqrt{6}} \times \$69,900,000$$



SQUARE ROOT LAW OF INVENTORY POOLING:



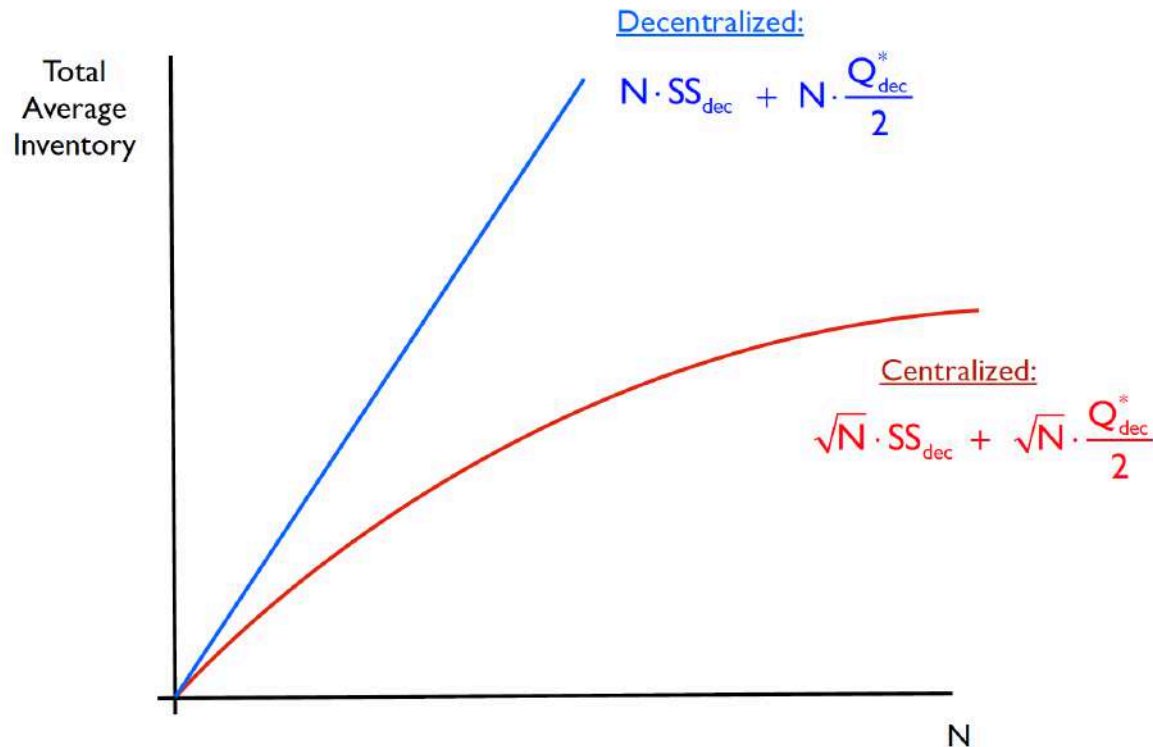
Sources of the square root efficiencies:

Average Inventory: $INV = \frac{1}{2} \sqrt{\frac{2\lambda s}{h}} + Z_{SL} \sigma \sqrt{L}$

Pooled Mean Demand: $\lambda(N) = N \times \lambda(1)$

Pooled Demand Variability: $\sigma(N) = \sqrt{N} \times \sigma(1)$

Square root law of pooling



Question:

In practice, consumer markets often exhibit correlated demand. The benefits of pooling increase as the correlation in demand:

- a) Increases
- b) Decreases

Forms of Pooling

- ✓ Common Location

Centralized warehouse

- ✓ Virtual Centralization

Linked by Information System and transship goods between locations

- ✓ Specialization

Each product stocked in only one location

- ✓ Component Commonality

One component fits all!

- ✓ Product Substitution

Red satisfies mauve and pink demand

- ✓ Postponement & Mass Customization

Stock generic, customize-to-order, delay product “localization”

Distribution Strategies

(Fundamentals of a delivery business)

There has never been a time of greater demand for last-mile transport



Urbanization

60%
people living in
cities in 2030

20–35%
congestion
increase
since 2010



Customers

2.1bn
people expected
to buy goods
online by 2021

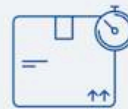
20%
online retail share
by 2023



Products

10%
per annum
e-grocery growth
worldwide

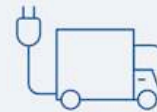
32%
of furniture sold
online by 2023 in
the US



Delivery

20%–40%
growth in
same-day

10%
per annum
growth in
instant delivery



Technology

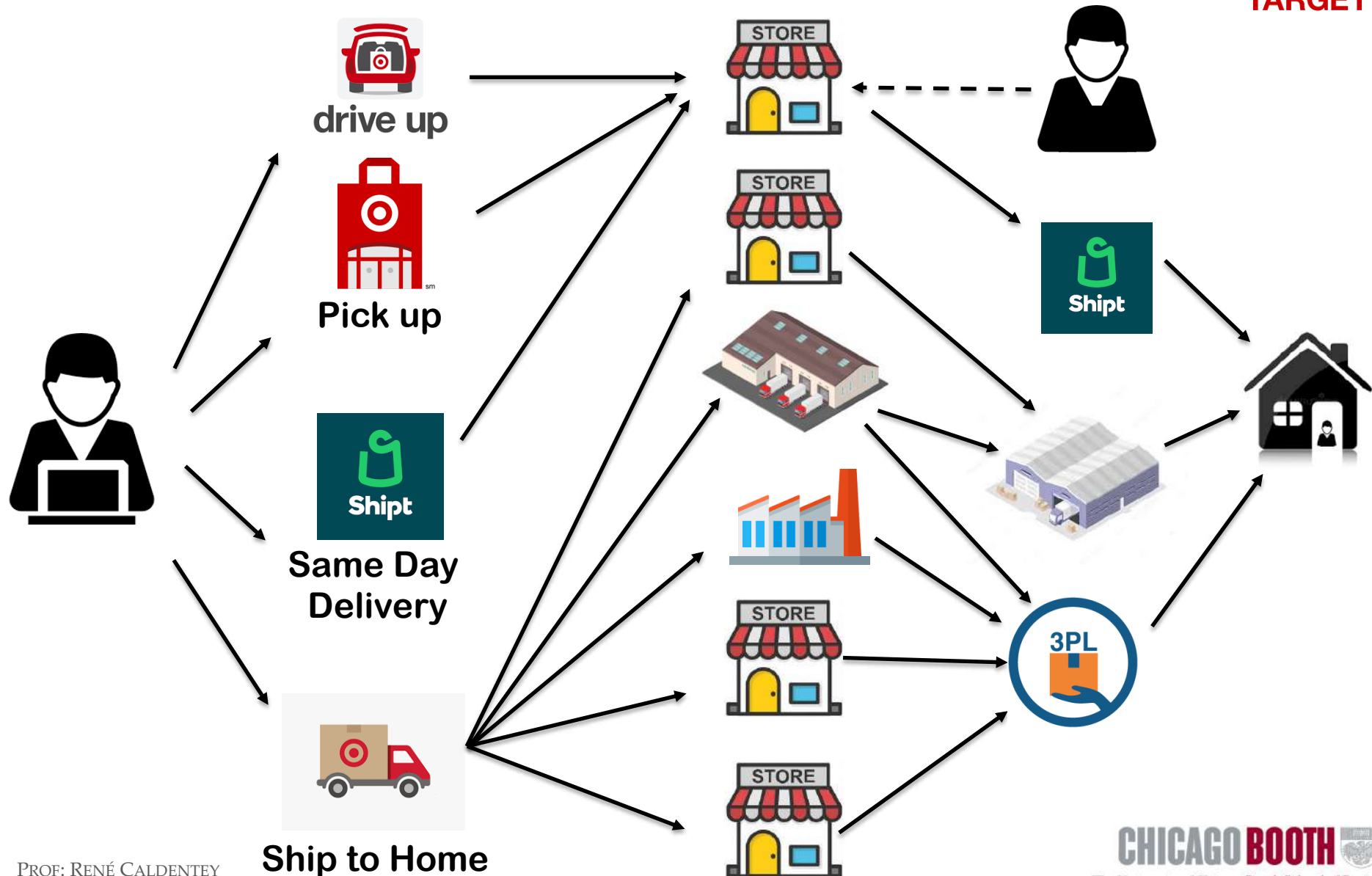
14–35%
xEV share of new
car sales across
regions by 2030

2024
year in which
most OEMs will
release L4/5
autonomous
vehicles

78%
growth through
2030 in urban last-mile
deliveries



Target Digital Fulfillment Network



Target opens first larger-format store in Katy

Shafaq Patel



More space: The new store layout delivers a backroom fulfillment space that's five times larger than previous stores of similar size. This additional space will help support the ongoing growth Target has experienced, with its stores fulfilling more than 95% of the retailer's digital orders and same-day services accounting for more than 10% of its overall sales.

The economics of last mile operations

Cost per order $\approx$ Picking/Packing + Last-mile + Failures/Returns



Picking/Packing Cost

~55% of warehouse OpEx

Source: de Koster et al., EJOR (2007).

Last-mile Cost

~41% of logistics/SC costs
~53% of shipping costs

Sources: Capgemini (2023);
Forbes Tech Council (2023).

Failures/Returns Cost

~\$10-\$20 excluding freight
Return rates: ~ 17.6% online
~ 10.02% B&M

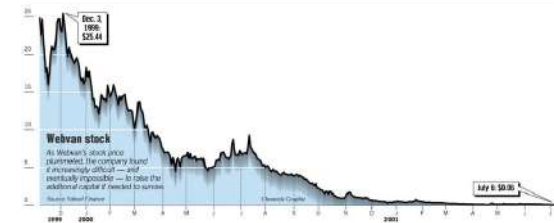
Sources: Brightpearl (2018);
Lian et al., MOR (2025).

Two competing e-distribution models



	Tesco.com	% of sale	Webvan	% of sale
Average Sale	\$123.25	100%	\$114	100%
Minus:				
Cost of Groceries	\$85.66	70%	\$83.00	73%
Other store costs	\$15.16	12%		
Marketing and admin.	\$5.67	5%	\$133.04	117%
Picking	\$8.50	7%	\$2.50 (est.)	2%
Delivery	\$9.30	7%	\$27.50 (est.)	24%
Plus:				
Delivery fee	\$7.25	6%	-	
Net Profit (Loss)	\$6.21	5%	(\$132.04)	

Source: Business Week Online 10/01/01



Traveling Salesman Problem (TSP)

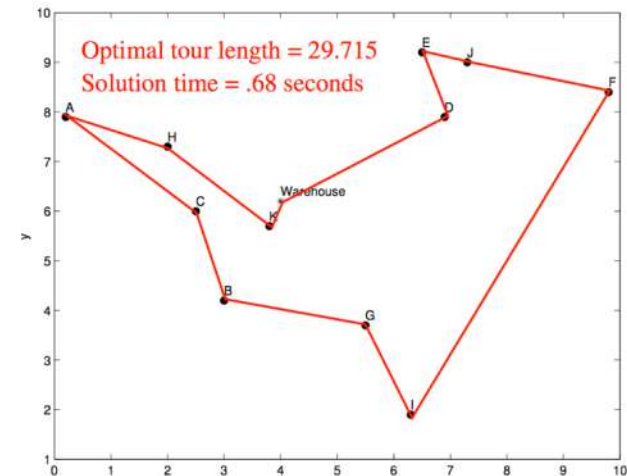
❖ Problem:

Find the shortest route between N “cities” where each city is visited exactly once and the route originates and terminates at a depot.

- ❖ First formulated mathematically in 1930s
- ❖ One of the most widely studied problem in OR
- ❖ The number possible routes grows exponentially
 - 5 customers: 120 possible routes
 - 10 customers: 3,628,800 possible routes
 - 20 customers: $2.43E+18$ possible routes
- ❖ Applications:
 - logistics and distribution
 - circuit board design
 - DNA sequencing, order sequencing

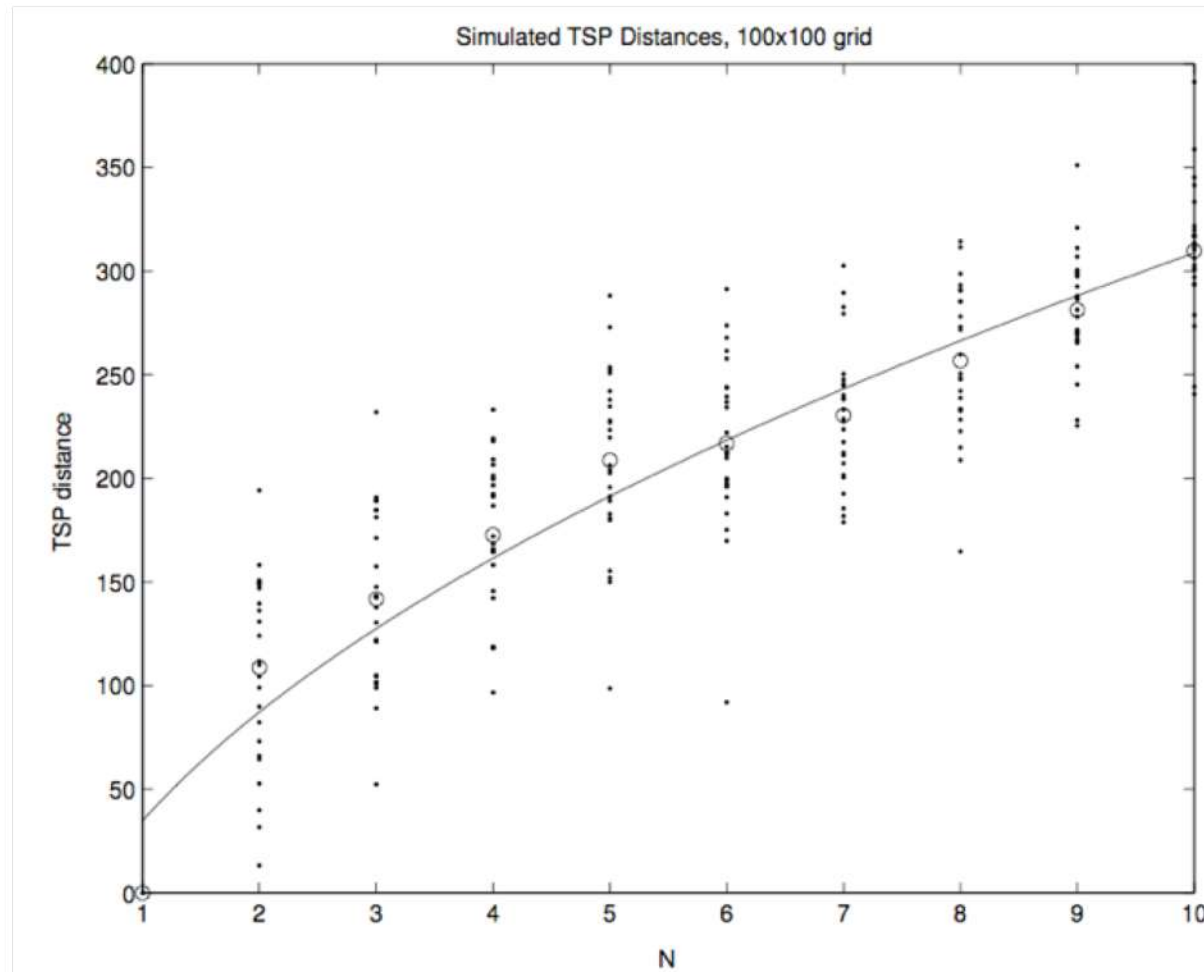


A poster from a contest run by Proctor and Gamble in 1962. The goal was to solve a 33 city instance of the TSP

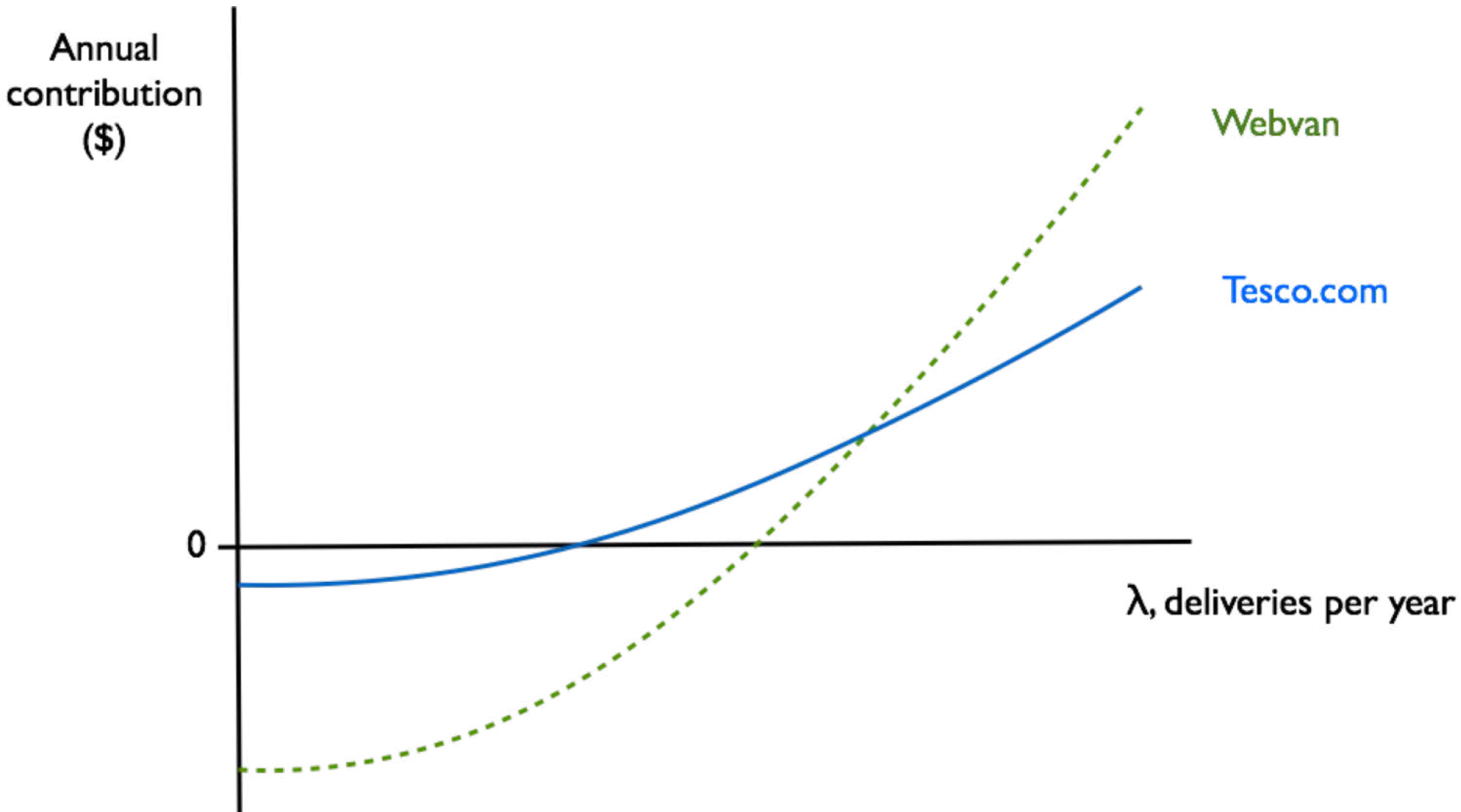


Steele's square root law of routing (1985)

Optimal TSP distance $\sim \sqrt{N}$, where N is the number of customers.

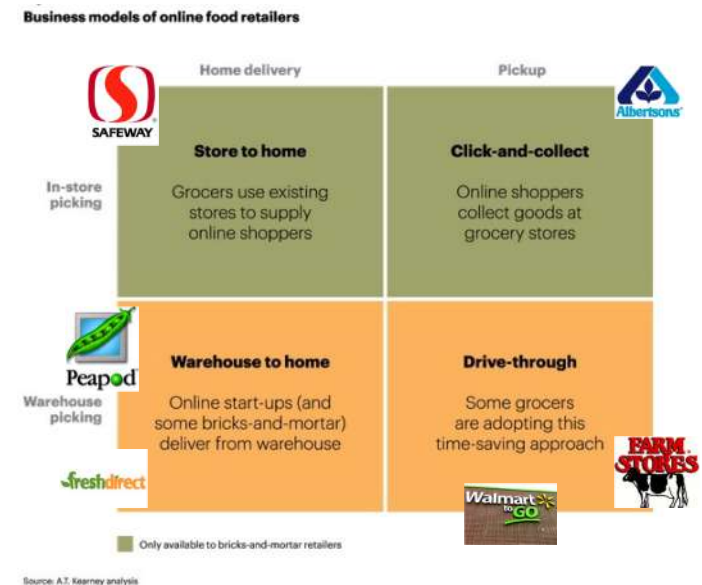


Webvan versus Tesco (conceptually)



Warehouse-pick versus store-pick

- ✧ Building a warehouse becomes more attractive with:
 - higher demand, λ
 - higher delivery densities
 - higher number of deliveries per route
 - higher delivery fee
 - wider time windows
 - closer to delivery zone
- ✧ Break-even point lower for store-pick
- ✧ Lose money until density increases
- ✧ Warehouse-pick more profitable than store-pick for high enough volumes



Module 8: Key Ideas

✓ Production Postponement

Delay product differentiating activities to cut mismatch inventory (risk pooling).

Quick response and mass customization

✓ Warehousing and Inventory Management

Distribution options: direct ship vs DC vs cross-dock = service–inventory–transport trade-off.

Network design: centralize vs localize DCs balances lead time vs inventory/fixed cost.

Square root law of inventory pooling: safety stock scales $\sim(1/\sqrt{N})$; correlation weakens benefits.

✓ Distribution Strategies

Fulfillment costs: picking/packing + last-mile + failures/returns.

Picking choice: warehouse-pick vs store-pick depends on volume and process efficiency.

Last-mile economics: density drives cost per stop; routing/time windows matter

Routing theory: VRP/TSP logic explains why consolidation and heuristics are essential.

Returns matter: reverse logistics and split shipments can erase margins—design for them

Square root law of routing

After Class Activities



GROUP:

WORK ON REPORTS AND PRESENTATION

SUBMIT YOUR FINAL PRESENTATION TONIGHT BY 10PM